

Aeronautical Systems : Sessions 1, 2, and 3

Advanced Aerospace Systems A

- 1 Nonlinear 6-DOF Motion Equations of a Rigid Body Expressed in Moving Coordinate System
- 2 Linearized Motion Equations
- 3 Fundamentals of State Equations and Linear Algebra
- 4 General Solution of State Equations and Mode Decomposition
- 5 Modeling Aerodynamic Forces (Fixed-Wing: Thin-Wing Theory, Strip Theory)
- 6 Stability Coefficients (Fixed-Wing Aircraft)
- 7 Motion Modes (Fixed-Wing Aircraft)
- 8 Summary

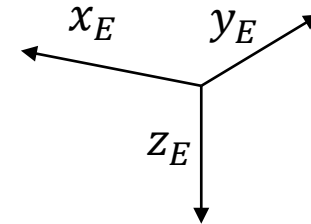
Advanced Aerospace Systems B

- 1 Fundamentals of Robust Control Theory 1
- 2 Fundamentals of Robust Control Theory 2
- 3 Fundamentals of Robust Control Theory 3
- 4 Stability Conditions and Feedback Control
- 5 Aerodynamic Modeling (Rotary Wing: Simple Momentum Theory, Wing Element Theory)
- 6 Stable Minority Coefficient (Multirotor Aircraft)
- 7 Motion Modes (Multirotor Aircraft)
- 8 Summary

1. Coordinate Systems
2. Euler Angles and Coordinate Transformations
3. State Variables Defining Aircraft Motion
4. Differential Equations in Moving Coordinate Systems
5. Nonlinear Equations of Motion

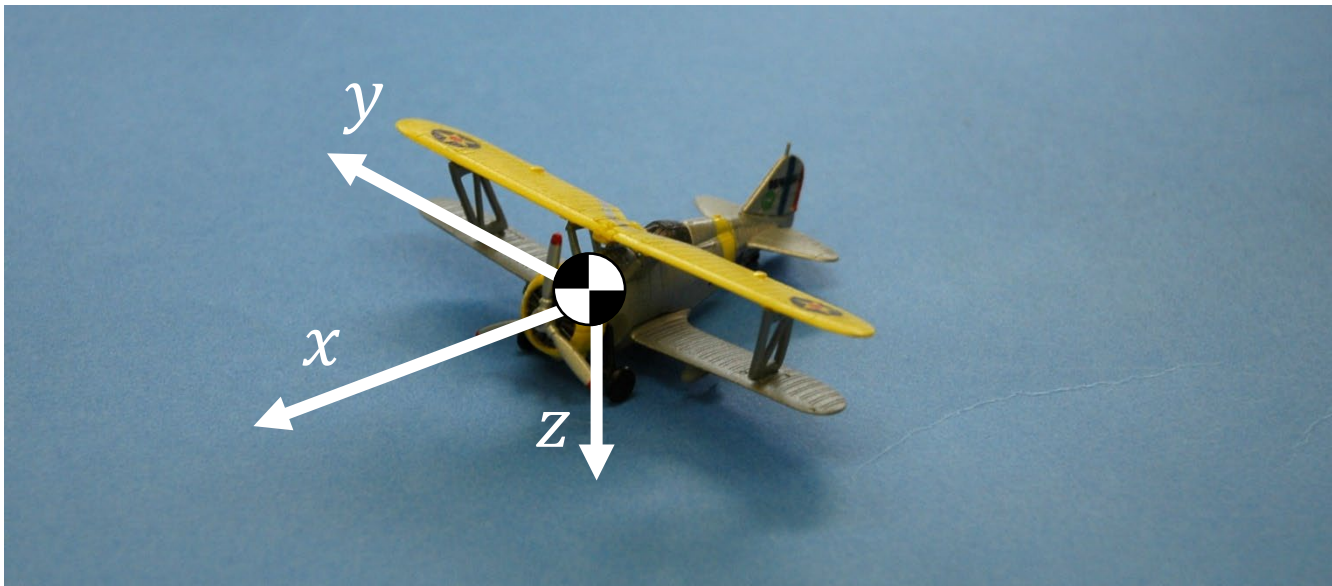
Ground-fixed coordinate system (inertial frame)

- Ground-fixed right-handed coordinate system
- Typically, z-axis pointing vertically downward



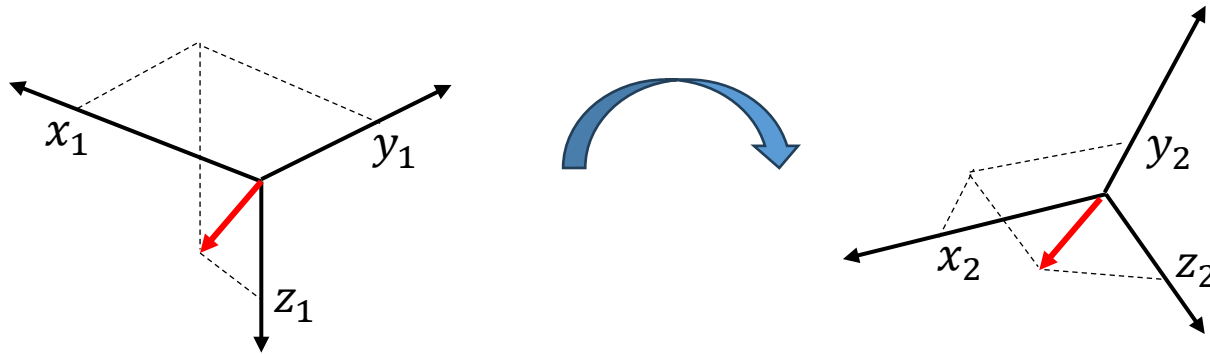
Aircraft-fixed coordinate system (moving coordinate system)

- Right-handed system fixed to the aircraft with origin at the center of gravity
- Typically, forward x-axis, rightward y-axis, downward z-axis



Coordinate Transformation

Find the relationship between the components of a fixed 3D vector in space represented in coordinate system 1 (x_1, y_1, z_1) and the components represented in the rotated coordinate system 2 (x_2, y_2, z_2).



$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = E \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \quad (\text{A1-1})$$

Coordinate Transformation

1. Rotate the coordinates $x_1 y_1 z_1$ Φ around the axis x_1 .

$$b_1 = \frac{z_2}{\cos \Phi} \quad b_2 = y_1 \tan \Phi$$

From $b_1 + b_2 = z_1$

$$z_2 = -y_1 \sin \Phi + z_1 \cos \Phi \quad \text{--- (#)}$$

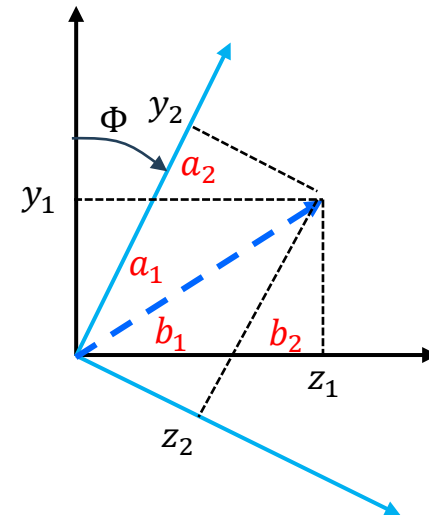
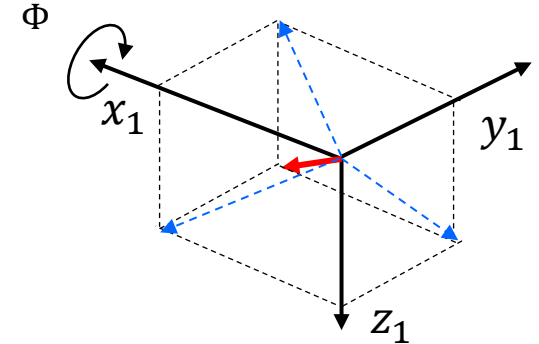
$$a_1 = \frac{y_1}{\cos \Phi} \quad a_2 = z_2 \tan \Phi$$

$a_1 + a_2 = y_2$ and from (#)

$$y_2 = y_1 \cos \Phi + z_1 \sin \Phi \quad \text{--- (##)}$$

(#), (##), and from $x_1 = x_2$

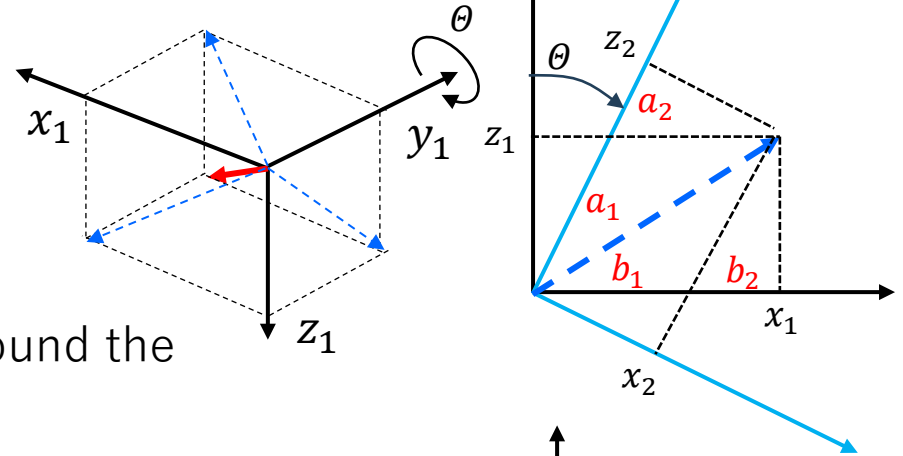
$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = E_x(\Phi) \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \quad E_x(\Phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \Phi & \sin \Phi \\ 0 & -\sin \Phi & \cos \Phi \end{bmatrix} \quad (\text{A1-2})$$



Coordinate Transformation

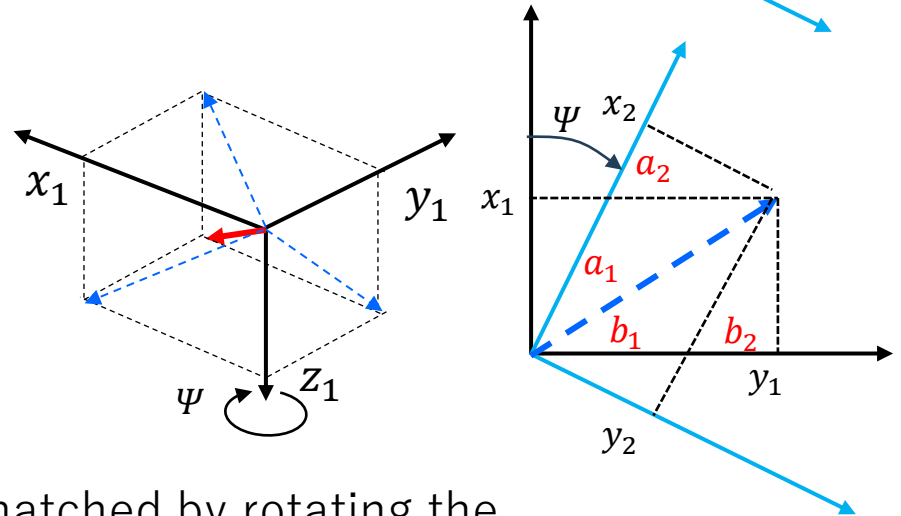
2. Rotate the coordinates $x_1y_1z_1$ θ around the axis

$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = E_y \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \quad E_y(\theta) = \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} \quad (\text{A1-3})$$



3. Rotate the coordinates $x_1y_1z_1$ ψ around the axis z_1 .

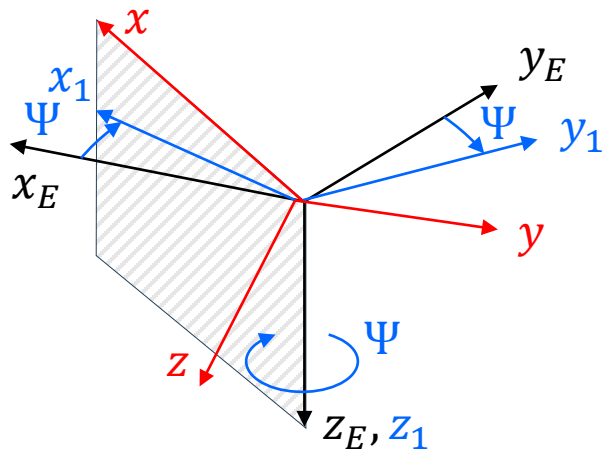
$$\begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix} = E_z \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} \quad E_z(\psi) = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{A1-4})$$



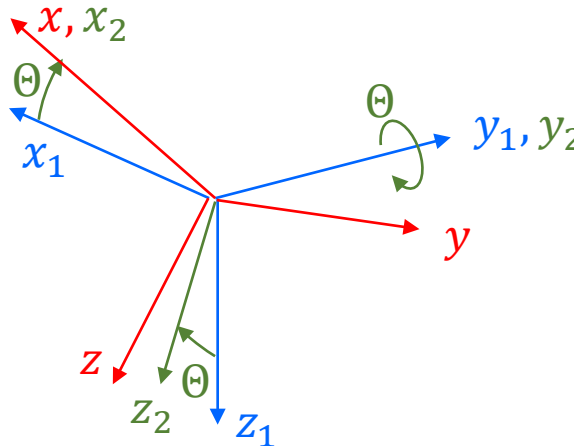
Generally, any coordinate $x_1y_1z_1$ can be matched by rotating the coordinate $x_2y_2z_2$ three times around each axis. There are 12 possible rotation sequences depending on the rotation axes and order. For aircraft, the attitude representation using the z axis, y axis, and x axis in that order (the 3-2-1 system) is used.

Euler angles: Three values defining the angle between the aircraft-fixed coordinate system (xyz) and the ground-fixed coordinate system ($x_E y_E z_E$)

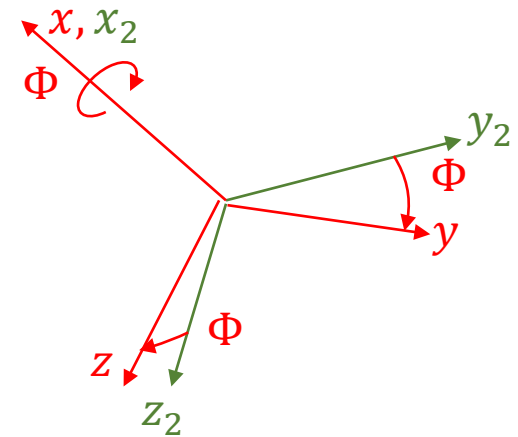
- ① Rotate the coordinate $x_E y_E z_E$ around the axis z_E so that the the axis x_E coincides with the $x - z_E$ plane. The rotation angle Ψ is called the azimuth angle or yaw angle, and the new coordinate is called the coordinate $x_2 y_2 z_2$.
- ② Rotate the coordinate $x_1 y_1 z_1$ around the axis y_1 so that the the axis x_1 coincides with the axis x . The rotation angle Θ is called the pitch angle, and the new coordinate $x_2 y_2 z_2$ is defined.
- ③ Rotate the coordinate $x_2 y_2 z_2$ around the axis x_2 to align it with the coordinate xyz . The rotation angle Φ called the roll angle or bank angle.



① Yaw angle



② Pitch angle

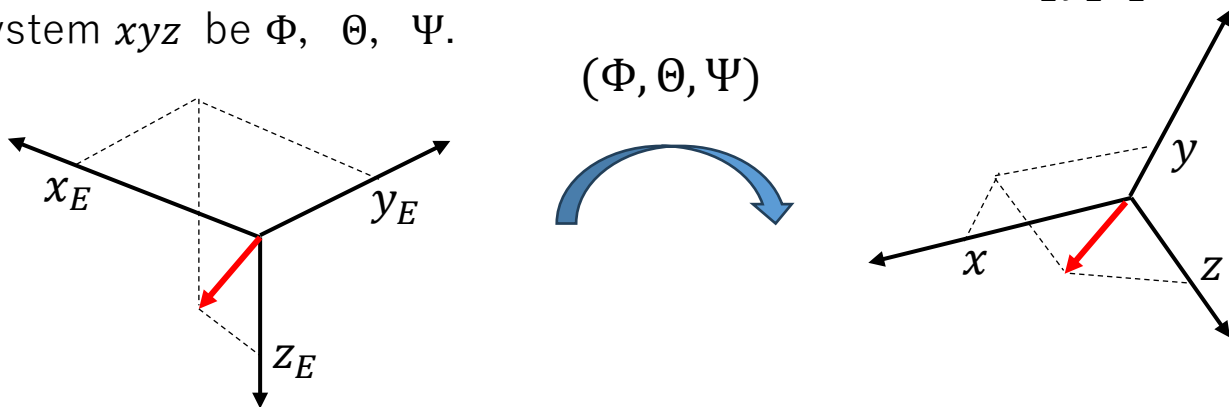


③ Roll angle

✂ Since rotation occurs in the order of z, y, x , these are called the 3-2-1 system Euler angles. Other definitions of Euler angles exist with different rotation sequences.

Coordinate Transformation

Let the 3-2-1 Euler angles from ground-fixed coordinate system $x_E y_E z_E$ to aircraft-fixed coordinate system $x y z$ be Φ , Θ , Ψ .



The components of a spatial fixed 3D vector represented in the ground fixed coordinate system (x_E, y_E, z_E) and in the aircraft fixed coordinate system (x, y, z) satisfy the following relationship.

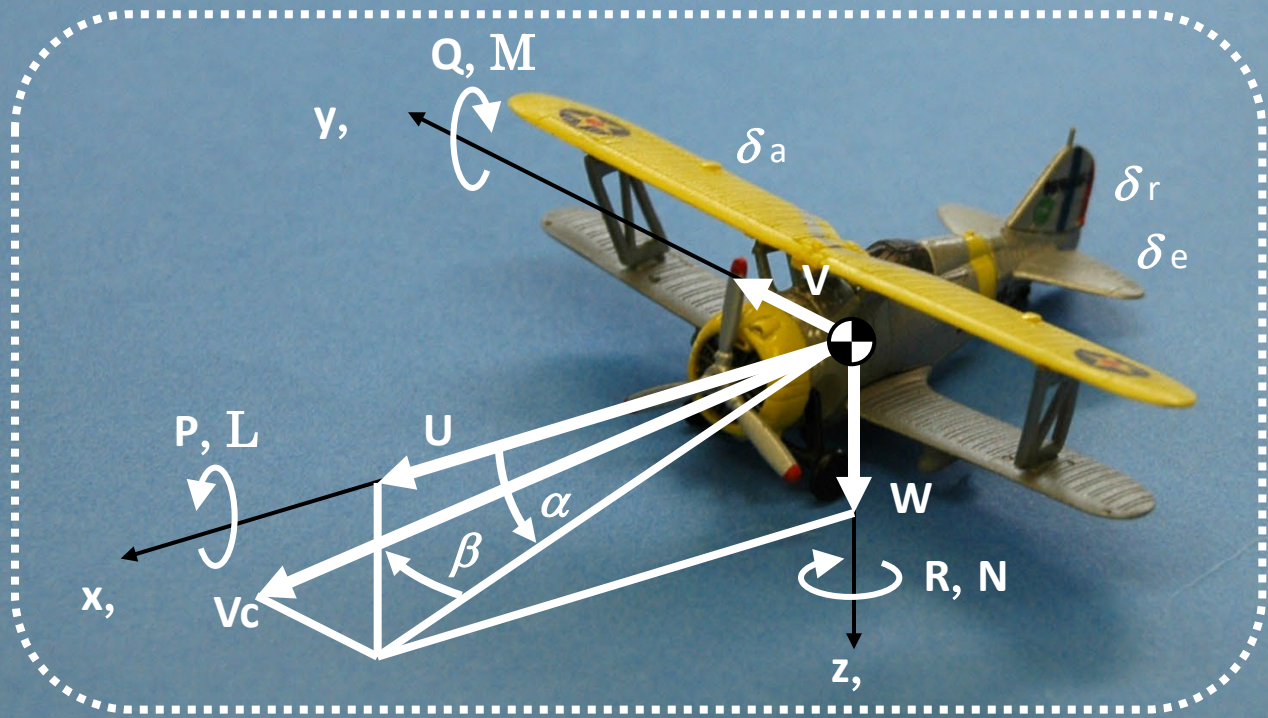
$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = E(\Phi, \Theta, \Psi) \begin{bmatrix} x_E \\ y_E \\ z_E \end{bmatrix} \quad (\text{A1-5})$$

$$E(\Phi, \Theta, \Psi) = \mathbf{E}_x(\Phi) \mathbf{E}_y(\Theta) \mathbf{E}_z(\Psi) \quad (\text{A1-6})$$

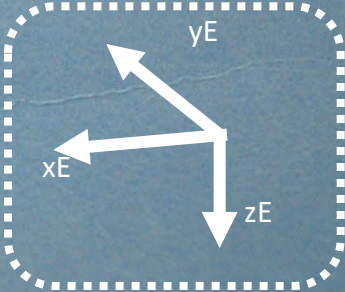
The matrix E is called the direction cosine matrix and can be obtained using Euler angles as follows.

$$\mathbf{E} = \begin{bmatrix} \cos \Theta \cos \Psi & \cos \Theta \sin \Psi & -\sin \Theta \\ \sin \Phi \sin \Theta \cos \Psi - \cos \Phi \sin \Psi & \sin \Phi \sin \Theta \sin \Psi + \cos \Phi \cos \Psi & \sin \Phi \cos \Theta \\ \cos \Phi \sin \Theta \cos \Psi + \sin \Phi \sin \Psi & \cos \Phi \sin \Theta \sin \Psi - \sin \Phi \cos \Psi & \cos \Phi \cos \Theta \end{bmatrix} \quad (\text{A1-7})$$

Airframe Fixed Coordinate System



Euler angles
(Φ, Θ, Ψ)
Ground-fixed coordinate system



x, y, z	Aircraft-fixed axis coordinate system	α	Angle of attack.
U, V, W	x, y, z axis direction velocity	β	Side slip angle.
P, Q, R	Angular velocity around the x, y, z axis	Φ, Θ, Ψ	Euler angle
F_x, F_y, F_z	External force in x, y, z axis direction	x_E, y_E, z_E	Ground-fixed coordinate system

Equation of Motion for Rigid Bodies (Newton's Second Law)

$$m \frac{d\mathbf{V}_c}{dt} = \mathbf{F} \quad (\text{A1-8a})$$

$$\frac{d\mathbf{h}}{dt} = \mathbf{G} \quad (\text{A1-8b})$$

$\mathbf{V}_c$:Velocity vector, $\mathbf{F}$:External force vector, $\mathbf{h}$:Angular momentum vector, $\mathbf{G}$:External force moment vector

Center of mass velocity vector

$$\mathbf{V}_c = U\mathbf{i} + V\mathbf{j} + W\mathbf{k} \quad (\text{A1-9a})$$

Angular velocity vector

$$\boldsymbol{\omega} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k} \quad (\text{A1-9b})$$

Position vector of infinitesimal element relative to center of mass

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k} \quad (\text{A1-10})$$

Definition
of center of gravity

$$\int \mathbf{r} dm = \mathbf{0} \quad (\text{A1-11})$$

$$\int dm = m \quad (\text{A1-12})$$

$\mathbf{i}, \mathbf{j}, \mathbf{k}$:Basis vectors of body-fixed x, y, z axis

Angular momentum vector about the center of mass

$$\begin{aligned}
 \mathbf{h} &= \int \mathbf{r} \times \mathbb{V} \, dm = \int \mathbf{r} \times \mathbf{V}_c \, dm + \int \mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r}) \, dm \\
 &= \left(\int \mathbf{r} \, dm \right) \times \mathbf{V}_c + \int (\mathbf{r} \cdot \mathbf{r}) \boldsymbol{\omega} \, dm - \int (\mathbf{r} \cdot \boldsymbol{\omega}) \mathbf{r} \, dm \quad \mathbf{V}_c \text{ is constant for each element} \\
 &= \int (x^2 + y^2 + z^2) \boldsymbol{\omega} \, dm - \int (xP + yQ + zR) \mathbf{r} \, dm \quad (\text{A1-13})
 \end{aligned}$$

Velocity vector of an infinitesimal element

$$\mathbb{V} = \mathbf{V}_c + (\boldsymbol{\omega} \times \mathbf{r}) \quad (\text{A1-14})$$

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c} \quad (\text{A1-15})$$

Substituting (A1-9b) and (A1-10) into (A1-13), we obtain;

$$\mathbf{h} = h_x \mathbf{i} + h_y \mathbf{j} + h_z \mathbf{k} \quad (\text{A1-16})$$

$$h_x = I_x P - I_{xy} Q - I_{xz} R \quad (\text{A1-17a})$$

$$h_y = -I_{yx} P + I_y Q - I_{yz} R \quad (\text{A1-17b})$$

$$h_z = -I_{zx} P - I_{zy} Q + I_z R \quad (\text{A1-17c})$$

Inertia

efficiency

$$I_x = \int (y^2 + z^2) dm, I_y = \int (z^2 + x^2) dm, I_z = \int (x^2 + y^2) dm \quad (\text{A1-18})$$

Inertia

product

$$I_{xy} = I_{yx} = \int xy dm, I_{xz} = \dots \quad (\text{A1-19})$$

Differentiation in moving coordinate systems

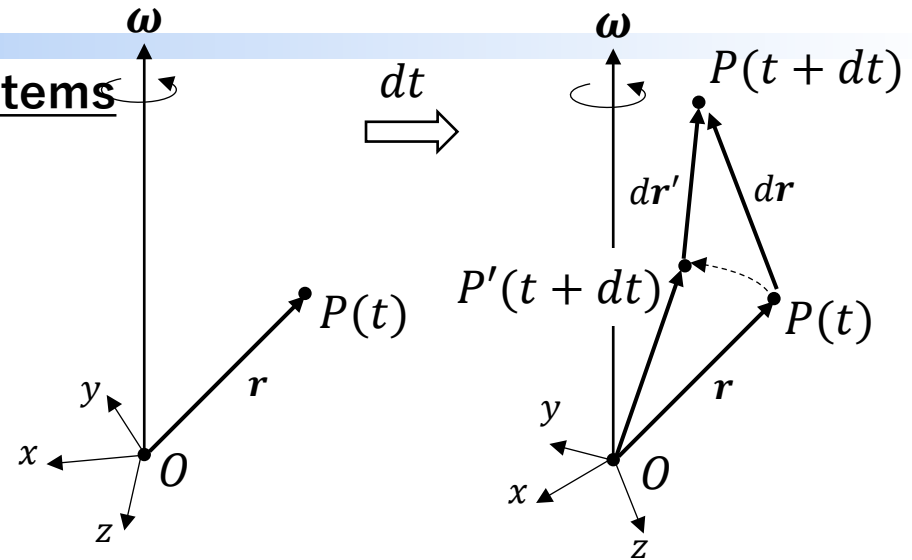
ω : Angular velocity of the moving coordinate axes

$\mathbf{r}$: Position vector of particle P

$d\mathbf{r}$: Displacement of particle P during dt

:

$d\mathbf{r}'$: Displacement of particle P during dt as seen by an observer rotating with the coordinate system



During dt , the coordinate system rotated at a speed of ω , and the point $P(t)$ moved to $P(t + dt)$. $P'(t + dt)$ is the point P that did not move when viewed by an observer rotating with the coordinate system

$$\overrightarrow{P(t)P(t + dt)} = \overrightarrow{P(t)P'(t + dt)} + \overrightarrow{P'(t + dt)P(t + dt)} \quad (\text{A1-20})$$

$$d\mathbf{r} = (\omega \times \mathbf{r})dt + d\mathbf{r}' \quad (\text{A1-21})$$

$$\frac{d\mathbf{r}}{dt} = (\omega \times \mathbf{r}) + \frac{d\mathbf{r}'}{dt} \quad (\text{A1-22})$$

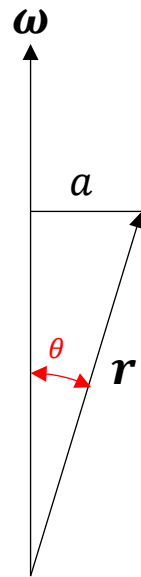
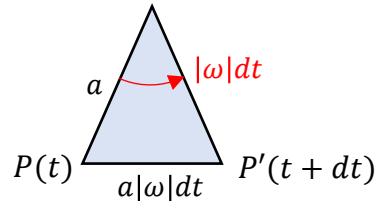
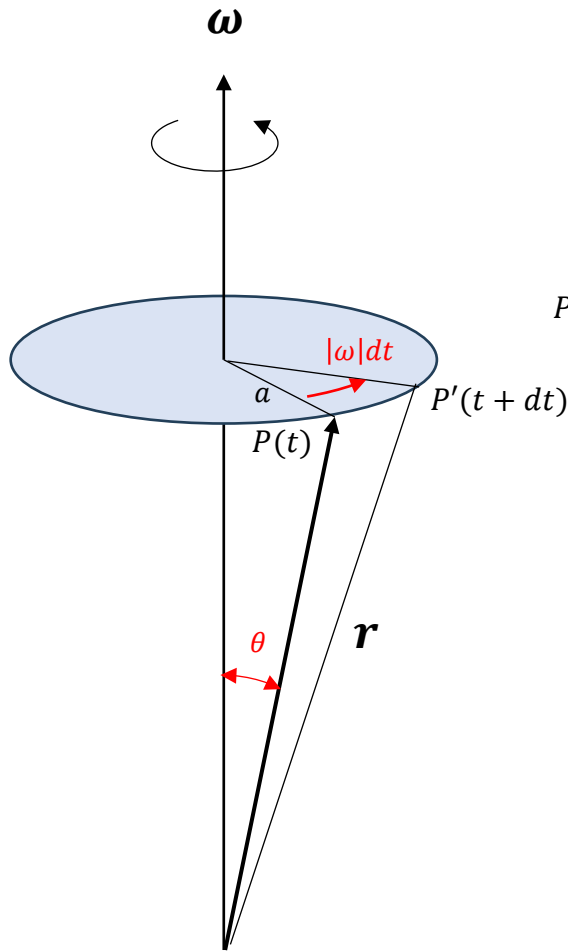
Rotation
of an axis

Change as seen from a rotating
coordinate system

Generally

$$\frac{d\mathbf{A}}{dt} = (\omega \times \mathbf{A}) + \frac{d^*\mathbf{A}}{dt} \quad \frac{d^*}{dt} : \text{Differentiation from the perspective of a moving coordinate system} \quad (\text{A1-23})$$

Nonlinear equations of motion (rigid body)



$$\overrightarrow{P(t)P'(t+dt)} = a|\omega|dt \quad (\text{A1-24})$$

$$a = |\mathbf{r}| \sin \theta \quad (\text{A1-25})$$

$$\left| \overrightarrow{P(t)P'(t+dt)} \right| = |\omega| |\mathbf{r}| \sin \theta dt \quad (\text{A1-26})$$

$\overrightarrow{P(t)P'(t+dt)}$ are orthogonal to ω and $\mathbf{r}$, so

$$\overrightarrow{P(t)P'(t+dt)} = (\omega \times \mathbf{r})dt \quad (\text{A1-27})$$

Nonlinear Equations of Motion (Rigid Body)

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = (\boldsymbol{\omega} \times \mathbf{r}) + \frac{d^*\mathbf{r}}{dt}$$

$$\frac{d\mathbf{A}}{dt} = (\boldsymbol{\omega} \times \mathbf{A}) + \frac{d^*\mathbf{A}}{dt}$$

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = (\boldsymbol{\omega} \times \mathbf{v}) + \frac{d^*\mathbf{v}}{dt} = \boldsymbol{\omega} \times \left((\boldsymbol{\omega} \times \mathbf{r}) + \frac{d^*\mathbf{r}}{dt} \right) + \frac{d^* \left((\boldsymbol{\omega} \times \mathbf{r}) + \frac{d^*\mathbf{r}}{dt} \right)}{dt}$$

$$= \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + \boldsymbol{\omega} \times \frac{d^*\mathbf{r}}{dt} + \frac{d^*\boldsymbol{\omega}}{dt} \times \mathbf{r} + \boldsymbol{\omega} \times \frac{d^*\mathbf{r}}{dt} + \frac{d^{*2}\mathbf{r}}{dt^2}$$

$$= \frac{d^{*2}\mathbf{r}}{dt^2} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + 2\boldsymbol{\omega} \times \frac{d^*\mathbf{r}}{dt} + \frac{d^*\boldsymbol{\omega}}{dt} \times \mathbf{r}$$

Acceleration as
seen from moving
coordinates

Centripetal
acceleration

Coriolis acceleration

[Example]

$m\mathbf{a} = \mathbf{F}$ Equations of motion in an inertial frame

$$m \left(\frac{d^{*2}\mathbf{r}}{dt^2} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + 2\boldsymbol{\omega} \times \frac{d^*\mathbf{r}}{dt} + \frac{d^*\boldsymbol{\omega}}{dt} \times \mathbf{r} \right) = \mathbf{F}$$

$$m \frac{d^{*2}\mathbf{r}}{dt^2} = -m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) - 2m\boldsymbol{\omega} \times \frac{d^*\mathbf{r}}{dt} - m \frac{d^*\boldsymbol{\omega}}{dt} \times \mathbf{r}$$

Equations of motion in
a moving coordinate
system

Centrifugal force (apparent force) Coriolis force (apparent force)

Nonlinear Equations of Motion (Rigid Body)

Equations of Motion in a Moving Coordinate System

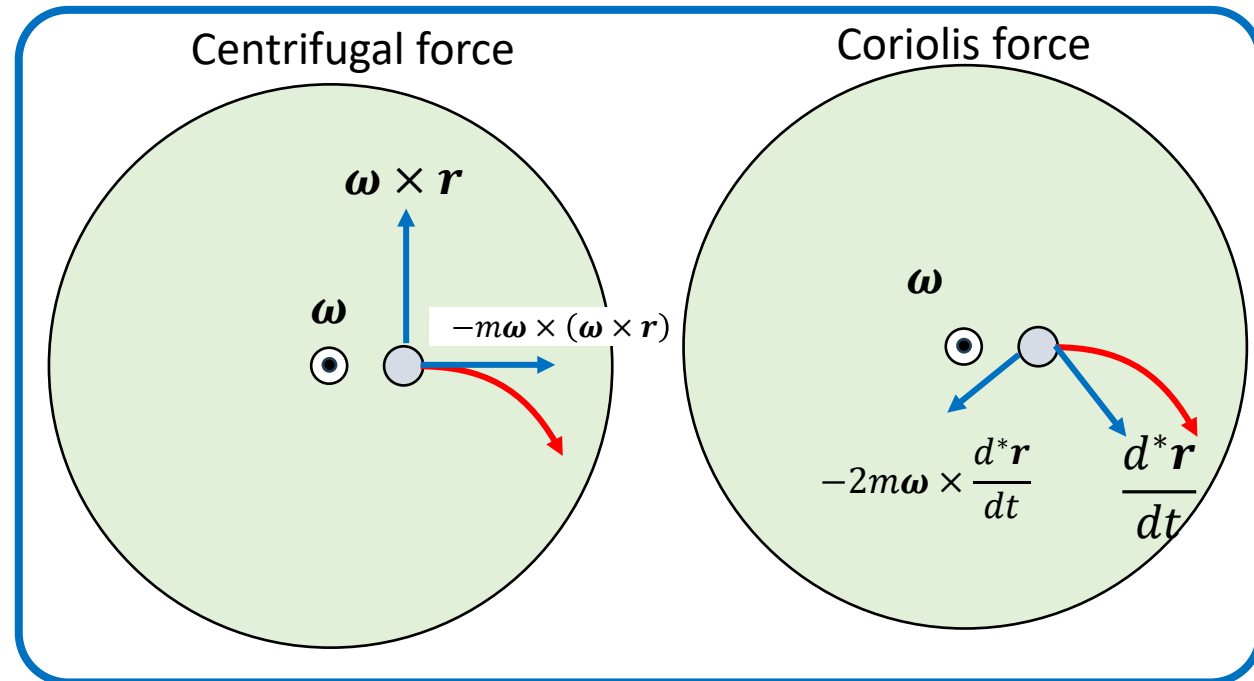
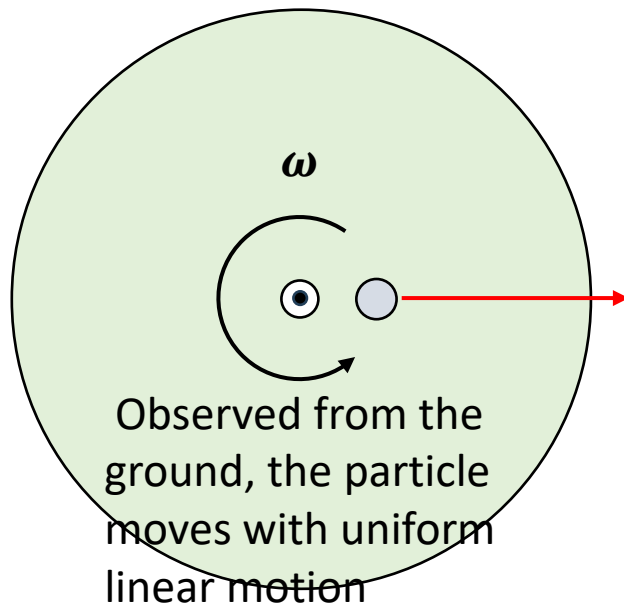
$$m \left(\frac{d^{*2} \mathbf{r}}{dt^2} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) + 2\boldsymbol{\omega} \times \frac{d^* \mathbf{r}}{dt} + \frac{d^* \boldsymbol{\omega}}{dt} \times \mathbf{r} \right) = \mathbf{F}$$

$$m \frac{d^{*2} \mathbf{r}}{dt^2} = \boxed{-m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})} - \boxed{2m\boldsymbol{\omega} \times \frac{d^* \mathbf{r}}{dt}} - m \frac{d^* \boldsymbol{\omega}}{dt} \times \mathbf{r} + \mathbf{F}$$

Centrifugal force
(apparent force)

Coriolis force (apparent force)

Giving initial velocity to a particle on a rotating disk



Path of the particle as observed by a person rotating with the disk